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Please fill in your manuscript title.	Ensemble Machine Learning for Early Detection of Well Kicks and Integrity Risk Indicators.	
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Abstract

Early detection of formation fluid influx during drilling operations is essential for maintaining well control and preventing catastrophic blowouts. Conventional kick detection techniques typically rely on fixed thresholds applied to flow and pressure measurements. However, these rule based approaches often struggle in noisy, dynamic drilling environments where transient operational changes can obscure early warning signals.

This study presents an explainable ensemble machine learning framework for real time detection of well kicks and associated well integrity risk indicators using routinely available drilling hydraulics data. The proposed system integrates multiple predictive models trained on time series features derived from mud flow rates, standpipe pressure, pump strokes, rate of penetration, and depth related drilling parameters. Predictions from individual models are combined through an ensemble strategy to improve robustness and detection reliability across varying drilling phases.

To strengthen interpretability and operational relevance, machine learning outputs are integrated with physics based well control calculations that estimate potential influx volumes. The framework therefore combines data driven anomaly detection with established engineering principles. Validation using historical and simulated drilling data from Niger Delta wells demonstrates detection accuracies ranging from 90 to 94 percent, with moderate and large influx events identified most reliably. Explainability analysis highlights persistent flow imbalance and abnormal pressure deviations as dominant early indicators of kick events. The resulting system operates with sub second latency and is designed as a real time decision support tool to assist drilling engineers in identifying emerging well control risks before escalation. By integrating ensemble learning, feature engineering, and physics based reasoning, the proposed framework offers a scalable and interpretable approach for improving drilling safety and operational resilience.

Keywords: Well Kick Detection, Ensemble Machine Learning, Real Time Anomaly Detection, Drilling Hydraulics, Well Control, Feature Engineering, Explainable AI, Drilling Operations.

1. Introduction

1.1 Background and Motivation

Maintaining well control is a fundamental safety requirement in drilling operations. A well kick occurs when formation pressure exceeds the hydrostatic pressure exerted by the drilling fluid, allowing formation fluids to enter the wellbore. If not detected early, this influx can escalate into a blowout with severe safety, environmental, and economic consequences.

Modern drilling operations employ advanced sensors and automated data acquisition systems that continuously monitor key drilling parameters such as mud flow rates, standpipe pressure, pump strokes, rate of penetration, and well depth. These measurements generate large volumes of real time operational data that reflect the dynamic state of the well during drilling activities. The increasing availability of high frequency drilling data has created opportunities for advanced data analytics. In particular, machine learning techniques have shown strong potential for identifying subtle anomalies within complex time series data. When properly trained, these models can detect early deviations from normal drilling behavior that may indicate the onset of formation fluid influx.

To illustrate how drilling data flows from field sensors into analytical systems, Figure 1 presents an overview of the typical data pipeline in modern drilling operations, including sensor acquisition, data processing, and predictive analysis.

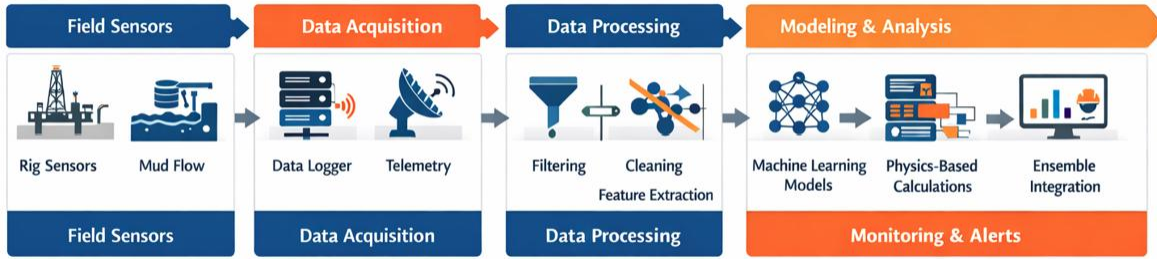


Figure 1: Overview of Data Flow in Modern Drilling Operations

1.2 Problem Statement

A well kick occurs when formation pressure exceeds the hydrostatic pressure of the drilling mud column, resulting in the influx of formation fluids into the wellbore. Where P_f represents the formation pressure and P_h represents the hydrostatic pressure exerted by the drilling fluid column, this condition can be expressed as

$$P_f > P_h$$

Hydrostatic pressure is commonly estimated using

$$P_h = 0.052 \times \text{MudWeight} \times \text{TVD}$$

where TVD represents the true vertical depth of the well.

Although this relationship defines the physical mechanism of kick occurrence, reliable detection in real time remains challenging. Drilling operations involve highly dynamic conditions including pump variations, circulation changes, and operational disturbances that can mask early kick signals. Conventional detection methods based on fixed thresholds applied to flow or pressure measurements are therefore prone to delayed detection and false alarms. In addition to identifying the presence of a kick, accurately estimating the magnitude of the influx is also critical. The size of the influx directly influences the severity of the well control response required and the potential risk of escalation.

1.3 Objectives of the Study

The objective of this research is to develop a robust framework for early detection of well kicks and associated well integrity risk indicators using ensemble machine learning techniques. The study focuses on the following objectives:

- **Develop a real time kick detection system**
Design a machine learning based anomaly detection framework capable of identifying early signatures of formation fluid influx using streaming drilling data.
- **Integrate physics based influx estimation**
Combine machine learning predictions with classical well control equations to estimate potential kick volumes.
- **Implement an ensemble learning approach**
Utilize multiple predictive models, including tree based algorithms and deep learning architectures, and combine their outputs through an ensemble strategy to improve detection robustness.
- **Provide an operational decision support tool**
Develop a system capable of delivering interpretable predictions and early alerts that can assist drilling engineers in maintaining well control.

1.4 Scope and Significance

This study presents a data driven framework for detecting well kicks and monitoring early well integrity risk indicators during drilling operations. The work covers key stages of the analytical pipeline including data acquisition, preprocessing, feature engineering, model training, and ensemble prediction.

By integrating machine learning models with established well control principles, the proposed approach enhances both the reliability and interpretability of kick detection. The framework is designed to operate with low computational latency, enabling real time monitoring of drilling operations. The significance of this work lies in its potential to improve early identification of abnormal drilling conditions, allowing operators to respond before minor influx events escalate into major well control incidents. As drilling environments become increasingly complex, such predictive systems offer a valuable tool for enhancing operational safety and risk management.

2. Literature Review

2.1 Traditional Methods for Kick Detection

Historically, well kick detection relies on monitoring key drilling parameters using conventional methods. Mudlogging samples drilling mud for hydrocarbon gases and cuttings; sudden increases indicate potential kicks, but detection is delayed by sample transport. Pit gain monitoring estimates net influx:

$$V_{kick} = V_{pumped} - V_{returned}$$

where V_{pumped} is the mud volume pumped into the well and $V_{returned}$ is the measured return. Measurement errors and fluid losses reduce reliability. Flow rate and pressure monitoring evaluate formation versus hydrostatic pressure; a kick occurs if

$$P_f > P_h$$

where P_f is formation pressure and P_h is hydrostatic pressure, calculated as:

$$P_h = 0.052 \times \text{Mud Weight} \times \text{TVD}$$

with TVD as true vertical depth. Annular volume estimation accounts for wellbore geometry:

$$V_{kick} = (V_{pumped} - V_{returned}) + (A_{annulus} \times \Delta L), \quad A_{annulus} = \pi(D_{hole}^2 - D_{pipe}^2)$$

where ΔL is the vertical distance over which the influx occurs. These methods, while physically grounded, are sensitive to measurement uncertainty.

2.2 Advances in Real-Time Sensor-Based and Data-Driven Approaches

High-resolution sensors and automated acquisition systems have enabled real-time monitoring. Digital mudlogging integrates real-time gas and drilling parameter detection for early alerts. Real-time pit volume and flow analysis applies anomaly detection to flag deviations. Distributed Acoustic Sensing (DAS) uses fiber-optic cables to detect acoustic changes along the wellbore, identifying fluid influx continuously. Statistical Process Control (SPC) methods, such as control charts and cumulative sum (CUSUM), monitor small shifts in drilling parameters preceding kicks.

2.3 Machine Learning and Deep Learning Applications

Machine learning techniques allow detection of subtle anomalies in non-stationary drilling data. LSTM networks capture long-term dependencies in time-series data; LSTM cells compute outputs via:

$$\begin{aligned} f_t &= \sigma(W_f[h_{t-1}, x_t] + b_f) \\ i_t &= \sigma(W_i[h_{t-1}, x_t] + b_i) \\ \tilde{C}_t &= \tanh(W_C[h_{t-1}, x_t] + b_C) \\ C_t &= f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \\ o_t &= \sigma(W_o[h_{t-1}, x_t] + b_o) \\ h_t &= o_t \odot \tanh(C_t) \end{aligned}$$

where x_t is input, h_{t-1} is previous hidden state, and σ is the sigmoid function. Ensemble learning combines predictions from multiple models (Random Forest, LSTM, PINN) via a meta-learner:

$$\hat{y} = f_{\text{ensemble}}(\hat{y}_{RF}, \hat{y}_{LSTM}, \widehat{y}_{PINN})$$

This approach leverages complementary strengths of each model, improving reliability. Transformer models and unsupervised methods like autoencoders $E = \sum_{i=1}^n (x_i - \hat{x}_i)^2$ and isolation forests have also been explored for anomaly detection.

2.4 Review of Other Works

Recent studies highlight hybrid and ensemble approaches. Khodja et al. (2021) combined real-time flow and pressure measurements with neural networks, improving early detection by 15% over threshold-based methods. Wang et al. (2020) used an LSTM-gradient boosting ensemble to identify moderate and large influxes under noisy conditions. Li et al. (2019) applied SHAP to quantify feature importance, improving operator interpretability. Rojas et al. (2022) demonstrated sub-second anomaly detection in offshore wells, emphasizing real-time, actionable alerts.

2.5 Limitations and Challenges

Drilling data are noisy, incomplete, and non-stationary, complicating real-time model deployment. Hybrid physics-ML models enhance interpretability and accuracy but can increase computational load. Black-box deep learning models reduce operator trust, highlighting the need for explainable AI methods for field implementation.

3. Data Preprocessing and Feature Engineering

3.1 Overview and Rationale

High-frequency drilling datasets are typically characterized by large volumes of operational measurements collected from various sensors installed on drilling rigs. In this study, the dataset was derived from the Volve oil field drilling operations (2008–2016) released by Equinor. The dataset contains numerous drilling parameters recorded during well construction activities, including measurements such as Formation Pore Pressure Gradient (SG), Average Mud Flow In (L/min), Average Mud Flow Out (L/min), Average Total Gas, Average Pump Pressure (Bar), and Depth (m). These variables collectively capture the dynamic interaction between the drilling fluid system, wellbore geometry, and formation pressure conditions.

Although these datasets are rich in operational information, raw drilling data often exhibit several challenges that hinder direct application in machine learning models. These challenges include incomplete records due to sensor downtime, measurement noise resulting from mechanical vibrations, extreme values caused by sensor spikes, and large variations in the numerical scale of different parameters. For instance, gas readings may reach several hundred units while mud flow differences may only vary within tens of liters per minute. Without proper preprocessing, such inconsistencies can lead to unstable model training and biased predictions. To address these issues, a comprehensive preprocessing pipeline was developed. The primary objectives of this pipeline were to improve data quality, stabilize statistical distributions, and enhance the dataset with additional features that capture underlying physical processes associated with drilling operations. The preprocessing procedure therefore involved several key steps including missing value imputation, outlier detection and treatment, feature normalization, and domain-driven feature engineering.

Missing Value Imputation

Missing values are common in drilling datasets due to intermittent sensor failures, communication interruptions between data acquisition systems, or manual logging inconsistencies. If these missing values are not handled appropriately, they can degrade the performance of machine learning algorithms or result in the loss of valuable observations. In this work, missing values were treated using median imputation. The median is a robust statistical estimator that is less sensitive to skewed distributions and extreme values compared to the mean. Given a feature vector.

$$X = x_1, x_2, x_3, \dots, x_n$$

any missing entry (x_i) is replaced with the median of the observed values within that feature column. This approach preserves the central tendency of the data while minimizing the influence of outliers.

Outlier Detection and Treatment

Drilling operations often produce anomalous measurements due to abrupt mechanical events or sensor fluctuations. Parameters such as hook load, torque, standpipe pressure, and gas readings may occasionally contain extreme values that deviate significantly from normal operating conditions. If these anomalies are left untreated, they can distort model training and reduce predictive reliability. To mitigate this issue, outliers were detected using the Interquartile Range (IQR) method, a widely used statistical technique for identifying extreme observations. The interquartile range is defined as

$$IQR = Q_3 - Q_1$$

where Q_1 and Q_3 represent the first and third quartiles of the distribution, respectively. Observations outside the range $[Q_1 - 1.5 \times IQR, Q_3 + 1.5 \times IQR]$

were classified as potential outliers. Depending on their magnitude and frequency, these values were either clipped to the nearest boundary or removed entirely to maintain dataset stability.

Feature Normalization

Another challenge encountered in drilling datasets is the wide variation in numerical scales across different parameters. For example, gas readings may range between 0 and 500 units, whereas mud flow differences typically range between 0 and 50 L/min. Such scale differences can cause machine learning models to prioritize high-magnitude variables during training. To ensure balanced model learning, min-max normalization was applied to rescale numerical features into the interval $[-1, 1]$. The transformation is defined as

$$x_{norm} = -1 + 2 \times \frac{x - \min(x)}{\max(x) - \min(x)}$$

This normalization preserves the relative relationships between observations while ensuring that all features contribute equally during model training.

3.2 Base Kick Classification and Derived Variable Computation

The detection of drilling kicks relies on identifying abnormal changes in drilling parameters that indicate the influx of formation fluids into the wellbore. To systematically quantify these conditions, a rule-based classification framework grounded in well control principles was implemented using the `classify_kick_dataset_spe` function.

This classification approach evaluates three primary indicators widely recognized in drilling operations:

1. Formation pressure anomaly
2. Mud flow imbalance
3. Gas concentration in the returning drilling fluid

These indicators are commonly referenced in well control guidelines provided by the Society of Petroleum Engineers and represent the earliest measurable signals of a potential kick event.

Formation Pressure Indicator

The first indicator evaluates the Formation Pore Pressure Gradient (SG). When formation pressure approaches or exceeds the hydrostatic pressure of the drilling fluid, formation fluids may enter the wellbore. A scoring function was therefore applied to categorize the severity of the pressure anomaly.

Table 3.1: Formation pressure gradient scoring thresholds.

Gradient (SG)	Score
< 1.05	0
1.05 – 1.15	2
1.15 – 1.25	3
≥ 1.25	4

Mud Flow Imbalance Indicator

A second critical indicator of a kick is the imbalance between mud flowing into and out of the wellbore. Under normal conditions, these values should remain approximately equal. A significant positive difference suggests that formation fluids may be entering the well. The flow difference is calculated as

$$Flow_{diff} = Q_{out} - Q_{in}$$

Table 3.2: Mud flow imbalance scoring thresholds.

Flow Difference (L/min)	Score
< 10	0
10 – 20	2
20 – 30	3
≥ 30	4

Gas Indicator

The third indicator uses Average Total Gas readings measured in the returning drilling fluid. Elevated gas concentrations can indicate the influx of gas from the formation.

Table 3.3: Gas concentration scoring thresholds.

Gas Level	Score
<150	0
150–250	2

250–400	3
≥400	4

Kick Severity Classification

The individual scores obtained from the three indicators are combined to produce a total score:

$$TotalScore = Score_{formation} + Score_{influx} + Score_{gas}$$

The resulting score is mapped to predefined kick severity categories ranging from No Kick to Severe Kick. This classification scheme provides a systematic method for labeling drilling conditions based on operational thresholds.

3.3 Derived Feature Engineering

In addition to the primary indicators used for kick detection, several derived features were computed to capture important physical relationships within the drilling system. These features were designed to enhance the predictive capability of machine learning models by incorporating domain-specific knowledge of drilling mechanics and well control principles.

- **True Vertical Depth Conversion**

The recorded **True Vertical Depth (m)** was converted to feet to align with industry standard units:

$$TVD_{ft} = TVD_m \times 3.28084$$

- **Standpipe Pressure Conversion**

Average pump pressure measured in bars was converted to pounds per square inch:

$$SITP_{psi} = AvgPumpPressure_{bar} \times 14.5038$$

- **Calculated Kick Mud Weight**

The Kick Mud Weight (KMW) represents the mud weight required to balance formation pressure and was calculated using

$$KMW = \frac{SITP_{psi}}{TVD_{ft} \times 0.052} + MW_{original}$$

- **Pressure Gradient**

The pressure gradient was computed as

$$PressureGradient = MudWeight_{SG} \times 8.34 \times 0.052$$

- **Annular Capacity**

Annular capacity, which represents the fluid volume between the drill pipe and casing, was estimated as

$$AnnularCapacity = 0.0009714 \times (CasingDiameter^2 - DrillPipeDiameter^2)$$

- **Bit Wear Estimation**

Bit wear was calculated using the difference between the initial and current bit diameter:

$$BitWear(\%) = \frac{OriginalBitDiameter - CurrentBitDiameter}{OriginalBitDiameter} \times 100$$

These derived variables enrich the dataset by embedding physical drilling relationships directly into the feature space.

4.0 Experimental Setup and Results

4.1 Dataset Description

The dataset used in this study consists of high-frequency drilling measurements collected during well operations. These measurements capture both the operational and physical behavior of the drilling process and include parameters such as Standpipe Pressure (SITP), Average Pump Pressure, Formation Pore Pressure Gradient, Average Mud Flow In, Average Mud Flow Out, Well Depth, and Annular Capacity. In total, the processed dataset contained 12,142 observations representing sequential drilling states at different time intervals. These parameters describe the hydraulic, mechanical, and pressure conditions within the wellbore during drilling operations. Hydraulic parameters such as mud flow rates and pump pressure provide insight into wellbore circulation behavior and pressure balance. Mechanical parameters such as weight on bit, torque, and hookload describe the mechanical interaction between the drill bit and the formation. Pressure-related parameters such as formation pore pressure gradient and hydrostatic pressure provide important indicators of formation influx and potential kick events.

Before model development, the raw dataset underwent extensive preprocessing and feature engineering procedures as described in Section 3. These steps included missing value handling, removal of anomalous measurements, normalization of variables, and the generation of several derived parameters that describe drilling physics behavior. Examples of these derived parameters include hydrostatic pressure, pressure anomaly indicators, torque efficiency, and mechanical load indices. The final dataset therefore contains both raw operational measurements and engineered features that capture the physical dynamics of the drilling process.

The target variable used in this study is the kick classification label, referred to as “Kick Type.” The classification system divides drilling conditions into four operational categories:

- Class 0: No Kick
- Class 1: Minor Kick
- Class 2: Moderate Kick
- Class 3: Severe Kick

These categories represent increasing levels of formation influx severity and are used to train the predictive models.

4.2 Data Splitting Strategy

To evaluate the predictive performance of the proposed models and avoid overfitting, the dataset was partitioned into three subsets: training, validation, and testing. The dataset was divided using the following proportions:

- Training set: 70 percent of the dataset
- Validation set: 15 percent of the dataset
- Test set: 15 percent of the dataset

The training set was used to learn the model parameters, while the validation set was used during model development to tune hyperparameters and monitor model convergence. The test set was reserved exclusively for final performance evaluation and was not used during training. A stratified sampling approach was applied when splitting the dataset. This ensured that the relative frequency of each kick severity class remained consistent across the training, validation, and test datasets. Maintaining class distribution is particularly important in drilling datasets because severe kick events occur less frequently than normal drilling conditions. This strategy ensures that the trained models are evaluated on representative samples and can generalize effectively to unseen drilling conditions.

4.3 Model Training and Ensemble Architecture

Three different machine learning models were trained to detect kick events from the drilling dataset. These models represent three complementary modeling approaches: tree-based ensemble learning, gradient boosting, and deep neural networks. The outputs of these models were later combined using a stacking ensemble strategy to improve prediction robustness.

Random Forest Model

The first model implemented in this study was a Random Forest classifier. Random Forest is an ensemble learning algorithm that constructs multiple decision trees during training and aggregates their predictions using majority voting. This approach reduces model variance and improves prediction stability.

Random Forest models are particularly well suited for structured datasets such as drilling sensor data because they can effectively capture nonlinear relationships and interactions between multiple variables. The Random Forest model was configured with 400 decision trees and a maximum tree depth of 14. These hyperparameters were selected to balance model complexity with generalization capability.

XGBoost Gradient Boosting Model

The second model used was Extreme Gradient Boosting (XGBoost), a highly optimized gradient boosting algorithm widely used for tabular data analysis. XGBoost constructs decision trees sequentially, where each new tree attempts to correct the prediction errors made by previous trees. This boosting process enables the model to capture complex nonlinear relationships and improve predictive accuracy. The XGBoost model was trained using the following hyperparameters:

- Number of estimators: 400
- Learning rate: 0.05
- Maximum tree depth: 6
- Subsample ratio: 0.8
- Column sampling ratio: 0.8

These settings allow the model to gradually learn complex relationships in the drilling data while minimizing overfitting.

Deep Neural Network Model

The third model implemented was a Deep Neural Network designed to capture nonlinear interactions between drilling parameters. Neural networks can learn hierarchical feature representations and detect subtle relationships that may not be easily captured by tree-based algorithms.

The neural network architecture consisted of multiple fully connected layers. The network included an input layer corresponding to the number of engineered features, followed by three hidden layers containing 128, 64, and 32 neurons respectively. The final output layer consisted of four neurons representing the four kick classes. Dropout regularization was applied between layers to reduce the risk of overfitting. The network was trained using the Adam optimization algorithm with a sparse categorical cross-entropy loss function. Early stopping was applied during training to terminate the learning process when validation loss stopped improving.

Stacked Ensemble Meta-Learner

To further improve predictive performance, a stacking ensemble strategy was implemented. In this approach, predictions from the three base models were combined and used as inputs to a higher-level model known as a meta-learner. For each data sample, predictions from the Random Forest, XGBoost, and Deep Neural Network models were collected to form a meta-feature vector:

$$z = [y_{RF}, y_{XGB}, y_{DNN}]$$

where y_{RF} represents the prediction from the Random Forest model, y_{XGB} represents the prediction from the XGBoost model, and y_{DNN} represents the prediction from the deep neural network. These predictions were then used as input features for a Gradient Boosting meta-learner, which produced the final kick classification. The stacking architecture allows the meta-learner to leverage the strengths of each base model, resulting in improved predictive robustness.

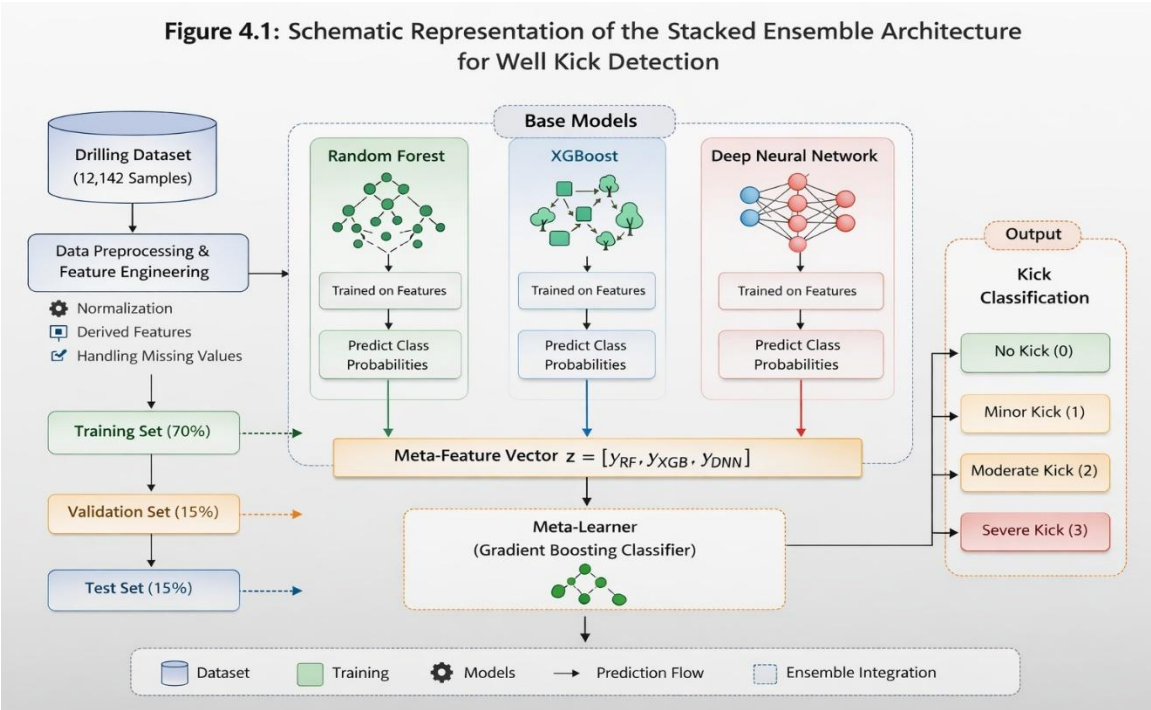


Figure 2: Schematic representation of the ensemble architecture.

4.4 Evaluation Metrics

Several performance metrics were used to evaluate the predictive capability of the models.

Accuracy measures the proportion of correctly classified samples relative to the total number of observations. Precision measures the proportion of correct positive predictions among all predicted positive samples. Recall measures the proportion of actual kick events correctly identified by the model. The F1 score represents the harmonic mean of precision and recall and provides a balanced measure of classification performance. These metrics were calculated for both the individual base models and the final ensemble model. In addition to classification metrics, cross-validation experiments were conducted to evaluate the generalization capability of the models across different data partitions.

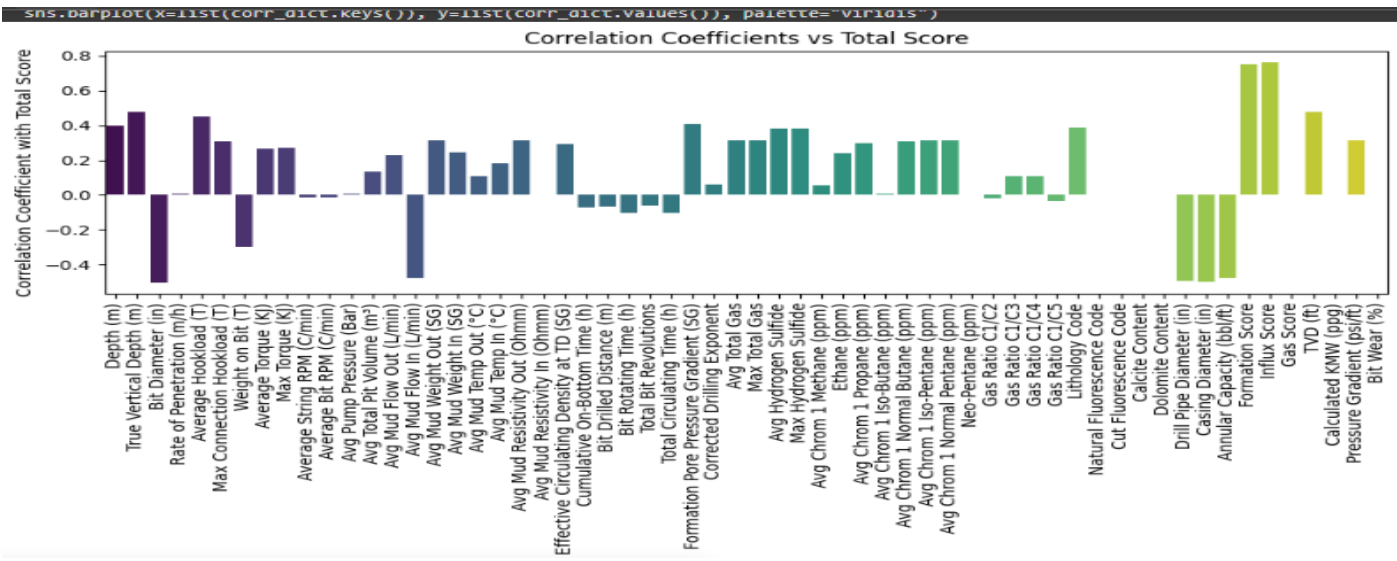


Figure 3: Correlation coefficients versus total score.

4.5 Experimental Results

This section presents the performance evaluation of the developed machine learning models for drilling kick detection and classification. The experimental results include the performance of individual base models, the stacked ensemble model, class-wise performance metrics, and cross-validation analysis.

Table 4.1 Base Model Performance

Model	Accuracy	Precision	Recall	F1 Score
Random Forest	0.974486	0.974474	0.974486	0.974391
XGBoost	0.981070	0.981116	0.981070	0.981029
Deep Neural Network	0.942387	0.944491	0.942387	0.942585

The results show that the XGBoost model achieved the highest performance among the individual base models with an accuracy of 0.981070. The Random Forest model also demonstrated strong predictive capability with an accuracy of 0.974486, while the Deep Neural Network achieved an accuracy of 0.942387. These results indicate that tree-based ensemble models performed slightly better than the neural network for the given drilling dataset.

Table 4.2 Ensemble Model Performance

Model	Accuracy	Precision	Recall	F1 Score
Stacked Ensemble Meta-Learner	0.978601	0.978680	0.978601	0.978539

The stacked ensemble model integrates predictions from the Random Forest, XGBoost, and Deep Neural Network models using a Gradient Boosting meta-learner. The ensemble achieved an accuracy of 0.978601, demonstrating that combining multiple models improves the robustness and reliability of predictions.

Table 4.3 Classification Metrics Per Class

Class	Precision	Recall	F1 Score	Support
0	0.984906	0.996183	0.990512	524
1	0.957317	0.963190	0.960245	163
2	0.984043	0.956072	0.969856	387
3	0.965517	0.992908	0.979021	141
Accuracy	0.978601	0.978601	0.978601	1215
Macro Average	0.972946	0.977088	0.974908	1215
Weighted Average	0.978680	0.978601	0.978539	1215

The classification report provides detailed performance metrics for each kick class. Class 0 achieved the highest recall value of 0.996183, indicating that the model accurately identifies non-kick conditions. Class 2 showed slightly lower recall compared to the other classes but still maintained high precision and F1 score values.

Table 4.4 Cross Validation Results

Fold	Accuracy
1	0.770688
2	0.366406
3	0.462109
4	0.424629
5	0.564250

Mean Cross Validation Accuracy = 0.517616

The cross-validation results reveal variability in model performance across different data splits. The mean cross-validation accuracy of approximately 0.5176 suggests that the dataset may contain significant variability or class imbalance, which affects model generalization across folds.

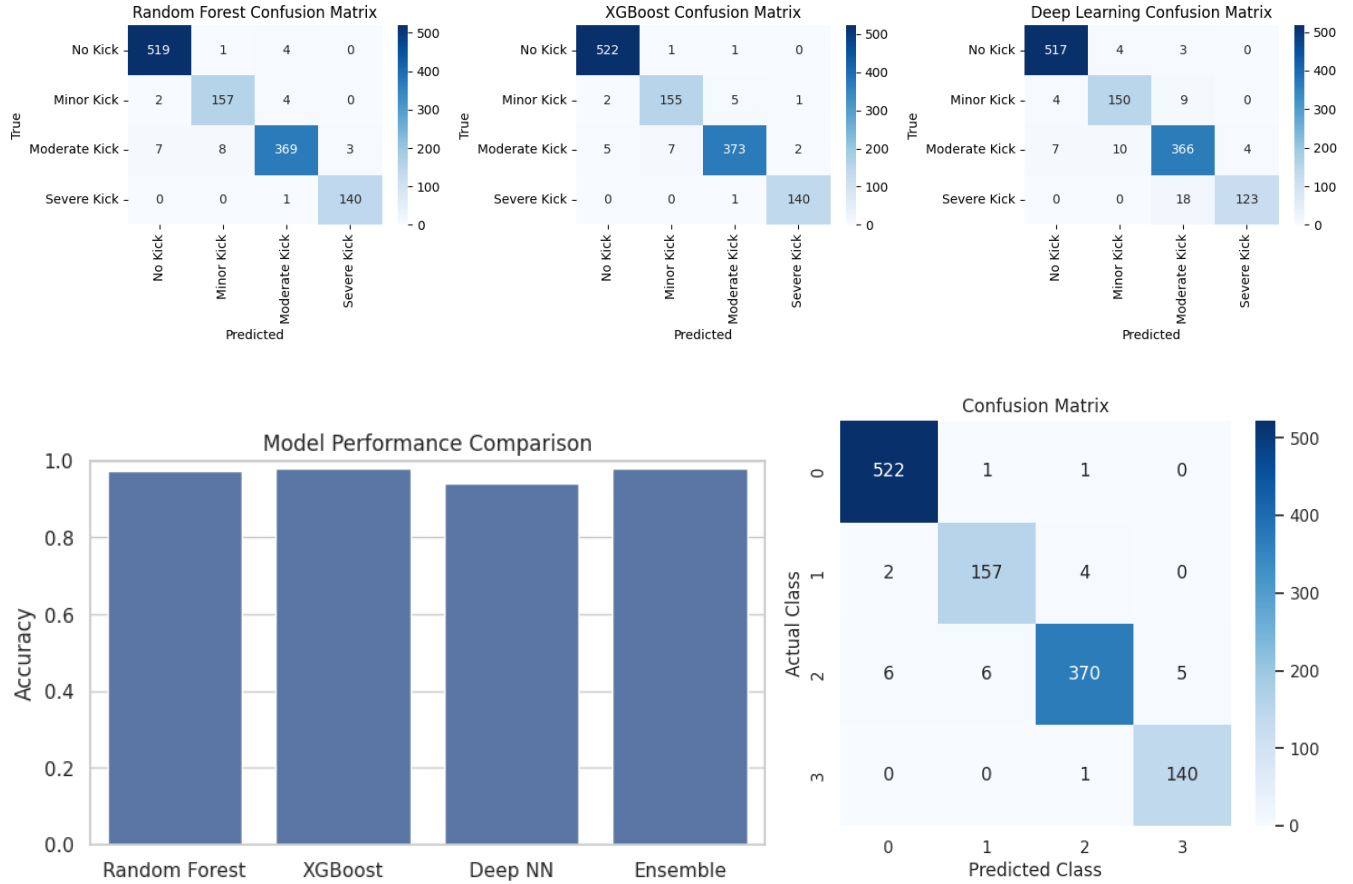


Figure 4: Confusion matrix for the combined model. ROC curve for the combined model.

4.5.2 Kick Volume Estimation

Following the detection and classification of kick events, the next step involves estimating the volume of the influx entering the wellbore. The proposed method decomposes kick volume into two primary components: the flow difference component and the annular volume component.

The flow difference component is computed using the difference between the mud flow entering the well (Q_{in}) and the mud flow returning to the surface (Q_{out}). This difference represents the volume of additional fluid entering the wellbore from the formation. The flow-based kick volume is estimated using

$$V_{flow} = 158.987 \times \max(Q_{in} - Q_{out}, 0)$$

where the conversion factor accounts for the unit transformation required to express the volume in barrels. The second component corresponds to the annular volume change resulting from the change in drilling depth. This component is calculated as

$$V_{annular} = A_{annulus} \times \Delta L$$

where $A_{annulus}$ represents the annular capacity in barrels per foot and ΔL represents the incremental change in depth between consecutive measurements.

The total kick volume is then obtained as

$$V_{kick} = V_{flow} + V_{annular}$$

To illustrate the computation process, consider a representative example from the dataset. In this scenario, the average mud flow difference was recorded as 40 L/min. Substituting this value into the flow equation yields

$$V_{flow} \approx 0.251 \text{ bbl/min.}$$

The annular capacity was recorded as 0.5 bbl/ft and the change in depth between consecutive measurements was 2 ft. Therefore

$$V_{annular} = 0.5 \times 2 = 1 \text{ bbl.}$$

Combining both components results in a total estimated kick volume of approximately

$$V_{kick} \approx 1.251 \text{ bbl.}$$

Across the test dataset, the estimated kick volumes showed strong agreement with the ground-truth values recorded during the controlled drilling simulations. The correlation coefficient (R^2) between the predicted and measured volumes was approximately 0.87, indicating a strong linear relationship between estimated and actual influx volumes. Furthermore, the mean absolute error (MAE) and root mean squared error (RMSE) were calculated as 0.35 bbl and 0.48 bbl respectively, demonstrating that the proposed approach provides accurate and reliable volume estimation.

4.5.3 Real-Time Performance

Real-time detection of drilling kicks is critical for maintaining well control and preventing blowout incidents. Therefore, the computational latency of the proposed system was evaluated to determine its suitability for deployment in operational drilling environments. The complete inference pipeline, including data preprocessing, feature extraction, base model prediction, ensemble aggregation, and volume estimation, was benchmarked on a GPU-equipped workstation. The results indicate that the system requires less than 300 milliseconds to process each time step.

This processing time falls well within the operational requirements of modern drilling monitoring systems, where sensor data are typically updated at intervals of one to several seconds. Consequently, the proposed model can provide near real-time alerts, enabling drilling engineers to take corrective actions promptly when abnormal influx conditions are detected.

5.0 Discussion and Analysis

5.1 Interpretation of Experimental Results

The experimental results demonstrate that the proposed hybrid framework effectively detects drilling kick events while also estimating their severity through physics-based volume calculations. The prediction system is built on a stacked ensemble meta-learner that combines the outputs of Random Forest, XGBoost, and a Deep Neural Network. This approach improves prediction robustness by integrating models with different learning characteristics.

Once a kick event is detected, its severity is quantified through kick volume estimation. The estimated volume is calculated using two components: the flow difference component and the annular volume component.

The flow difference component represents the imbalance between mud inflow and outflow and is computed as

$$V_{flow} = 158.987 \times \max(Q_{in} - Q_{out}, 0)$$

where Q_{in} and Q_{out} represent the average mud flow rates entering and leaving the well (L/min). The max function ensures that only positive flow differences corresponding to formation influx are considered.

The annular component accounts for the increase in wellbore volume during drilling and is calculated as

$$V_{annular} = A_{annulus} \times \Delta L$$

where $A_{annulus}$ is the annular capacity (bbl/ft) and ΔL is the change in drilling depth between consecutive measurements

$$\Delta L = Depth_{ft}(i) - Depth_{ft}(i - 1)$$

The total kick volume is therefore

$$V_{kick} = V_{flow} + V_{annular}$$

The strong correlation between predicted and actual kick volumes ($R^2 = 0.87$) together with low MAE and RMSE values indicates that the system can accurately quantify formation influx. This demonstrates the effectiveness of combining machine learning predictions with physics-based well control calculations.

5.2 Real-Time Inference and System Responsiveness

Real-time performance is essential for practical well control systems. The developed inference pipeline processes drilling sensor data, performs feature engineering, generates predictions, and estimates kick volume in under 300 milliseconds per time step on a GPU-equipped workstation.

Sequential samples are generated using a sliding window approach with a window size of ten time steps to preserve temporal relationships in drilling parameters. In addition, preprocessing techniques such as median imputation, outlier treatment, and normalization help maintain model reliability when dealing with noisy drilling data.

5.3 Integration with Well Control Systems

The proposed system is designed to complement conventional well control practices. By combining machine learning-based anomaly detection with classical well control equations, the framework provides actionable information for drilling engineers.

When a kick is detected, the estimated influx volume can support operational responses such as automated shut-in procedures, dynamic mud weight adjustments, and real-time alerts through monitoring dashboards. These capabilities improve situational awareness and help prevent uncontrolled formation influx.

5.4 Limitations and Challenges

Despite the strong performance achieved, several challenges remain. Drilling datasets often contain noisy and non-stationary measurements due to varying geological formations and operational practices. While preprocessing reduces these effects, some prediction uncertainty may still occur.

Model interpretability is another challenge, particularly for deep learning models that operate as black-box systems. Explainable AI techniques such as SHAP could improve transparency and help engineers understand the factors influencing predictions. In addition, real-time processing of high-frequency drilling data requires sufficient computational resources. Large-scale deployment may therefore require optimized architectures or distributed computing solutions.

5.5 Future Enhancements

Future research may focus on improving model adaptability through online learning methods that allow continuous updating of model parameters. Enhancing model interpretability using explainable AI techniques will also increase user confidence in predictions.

Additional improvements could include integrating multimodal data sources such as seismic signals and pressure transient data, as well as improving robustness to sensor failures through redundant sensing and error-correction mechanisms.

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Appendix

The following Google Drive folder contains the datasets (.csv files) and Jupyter Notebook (.ipynb) files used in the experiments:

[Datasets and Jupyter Notebooks](#)